Exercise sheet: Week 2

1. Find the likelihood, log-likelihood and MLE for a random sample $\underline{x} = (x_1, ..., x_n)$ from an exponential distribution for two different parameterisations $f(x; \lambda) = \lambda e^{-\lambda x}$ and $f(x; \theta) = \frac{1}{\theta} e^{-\frac{x}{\theta}}$. Compare your results with the method of moments estimator.

2. Find the likelihood, log-likelihood and MLE for a random sample $\underline{x} = (x_1, ..., x_n)$ from a Beta$(\alpha, 1)$ distribution with p.d.f. $f(x; \alpha) = \alpha x^{\alpha-1}$, $0 < x < 1$. What kinds of values must the x_i take if the MLE of α takes a very large value? Is the MLE for this example the same as that arising from the method of moments?

3. Identify sufficient statistics for the model parameters for the following random samples.

   (a) A random sample of size n from a Gamma(2, β) distribution.

   (b) A random sample of size n from a Beta(1, α) distribution.

4. A sample of claim sizes $x_1, ..., x_n$ is believed to be a random sample from a Pareto distribution with p.d.f. $f(x, \theta) = \frac{\theta}{(1+x)^{\theta+1}}$, $0 < x < \infty$, $\theta > 0$.

   (a) Determine the likelihood function $L(\theta; \underline{x})$ and show that $\sum \log(1 + X_i)$ is sufficient for θ. Find the MLE for θ.

   (b) Let $Y_i = \log(1 + X_i)$. Show that the Y_i have an Exp$(\frac{1}{\mu})$ distribution where $\mu = \frac{1}{\theta}$ and write down the MLE of μ. Compare this with your result from (a). Which property of the MLE does this illustrate?

5. Suppose that you carry out two independent binomial experiments with n_1 and n_2 trials in each and respectively x_1 and x_2 successes.

   (a) Suppose that the probability of success p is the same for each experiment. Find the MLE of p based on the 2 experiments jointly.

   (b) Suppose that the probability of success is respectively p_1 and p_2 in the two experiments where $0 < p_1, p_2 < 1$. Find the MLE of (p_1, p_2).

   (c) What is the MLE of (p_1, p_2) if we impose the condition that $p_2 \geq p_1$? *You will have two situations to consider depending on the values of* $\frac{x_1}{n_1}$ *and* $\frac{x_2}{n_2}$. Can you think of a practical situation in which the condition $(p_2 \geq p_1)$ will arise?